

ECON 3201

Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

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Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- i. An “Index” or “Home” page. This page should include the following:
 - Your name
 - Your photo (optional)
 - Your credentials
 - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]
- ii. A blog page. This will be the landing site for future assignments [5 Marks]
- iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- iv. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

Paste your link here

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

A valid probability distribution must have non-negative probabilities and the total probability must equal 1.

$$0 \leq \frac{1}{12} \leq 1$$

$$\sum_{x=1}^{12} f_X(x) = 12 \times \frac{1}{12} = 1$$

Therefore, this is a valid probability distribution.

(b) Compute the expected value of X .

The expected value is calculated using:

$$E(X) = \sum_{x=1}^{12} xP(X = x)$$

Since each value has a probability of $1/12$:

$$E(X) = \frac{1 + 2 + 3 + \dots + 12}{12}$$

$$E(X) = \frac{78}{12} = 6.5$$

Therefore, the expected value of X is 6.5.

(c) Compute the variance of X .

The variance is calculated using:

$$Var(X) = E(X^2) - [E(X)]^2$$

First, calculate $E(X^2)$:

$$E(X^2) = \frac{1^2 + 2^2 + 3^2 + \dots + 12^2}{12}$$

$$E(X^2) = \frac{650}{12} = 54.1667$$

Now calculate the variance:

$$Var(X) = 54.1667 - (6.5)^2$$

$$Var(X) = 54.1667 - 42.25$$

$$Var(X) = 11.9167$$

Therefore, the variance of X is approximately 11.9167.

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

Given $X \sim N(4, 5)$, the mean is 4 and the variance is 5.
The standard deviation is:

$$\sigma = \sqrt{5} = 2.236$$

Using the Z-score formula:

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{6 - 4}{\sqrt{5}} = 0.8944$$

Therefore:

$$P(X \leq 6) = P(Z \leq 0.8944)$$

$$P(X \leq 6) \approx 0.8145$$

(b) $Pr(X > 4)$

Given $X \sim N(4,5)$, the mean is 4 and the variance is 5.
Using the Z-score formula:

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{4 - 4}{\sqrt{5}} = 0$$

Since the normal distribution is symmetric around its mean:

$$P(X > 4) = P(Z > 0)$$

$$P(X > 4) = 1 - 0.5 = 0.5$$

Therefore, the probability is 0.5 or 50%.

(c) $Pr(-1 \leq X < 1)$

Given $X \sim N(4,5)$, the mean is 4 and the standard deviation is the square root of 5.
Using the Z-score formula:

$$Z = \frac{X - \mu}{\sigma}$$

For $X = -1$:

$$Z_1 = \frac{-1 - 4}{\sqrt{5}} = -2.2361$$

For $X = 1$:

$$Z_2 = \frac{1 - 4}{\sqrt{5}} = -1.3416$$

Therefore:

$$P(-1 \leq X < 1) = P(-2.2361 \leq Z < -1.3416)$$

$$= P(Z < -1.3416) - P(Z < -2.2361)$$

$$= 0.0899 - 0.0127$$

$$= 0.0772$$

Therefore, the probability is approximately 0.0772 or 7.72%.

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

Given $Z \sim N(0,1)$, the mean is 0 and the standard deviation is 1. Since Z is already standardized, we can use the standard normal distribution table directly.

$$P(Z \leq 0.83) = \Phi(0.83)$$

$$P(Z \leq 0.83) = 0.7967$$

Therefore, the probability is approximately 0.7967 or 79.67%.

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

Given $Z \sim N(0,1)$, we need to find the probability between -0.83 and 0.83. Using the standard normal distribution:

$$P(-0.83 \leq Z \leq 0.83) = \Phi(0.83) - \Phi(-0.83)$$

From the standard normal table:

$$\Phi(0.83) = 0.7967$$

Since the distribution is symmetric:

$$\Phi(-0.83) = 1 - 0.7967 = 0.2033$$

Therefore:

$$P(-0.83 \leq Z \leq 0.83) = 0.7967 - 0.2033$$

$$= 0.5934$$

The probability is approximately 0.5934 or 59.34%.

R Exercises

Q5

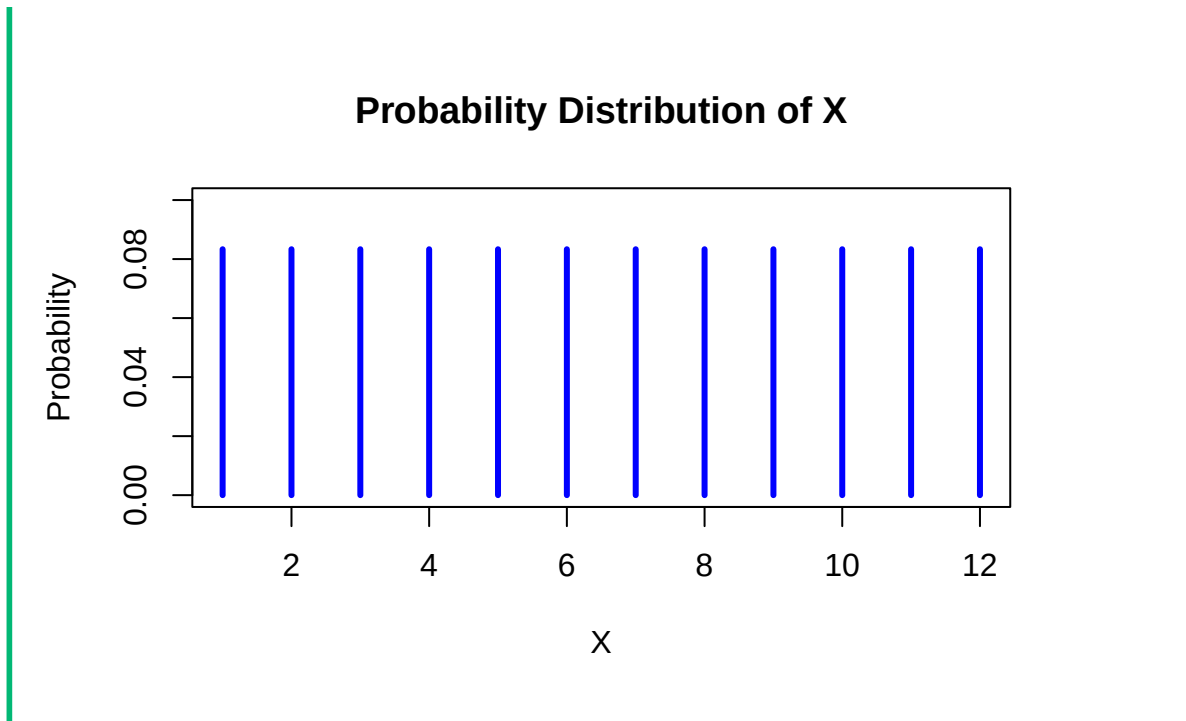
Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

- (a) Plot the probability distribution function $f_X(x)$ in R.

```
# Values of X
x <- 1:12

# Probability of each value
prob <- rep(1/12, 12)

# Plot the probability distribution
plot(x, prob,
     type = "h",
     main = "Probability Distribution of X",
     xlab = "X",
     ylab = "Probability",
     col = "blue",
     lwd = 3,
     ylim = c(0, 0.1))
```



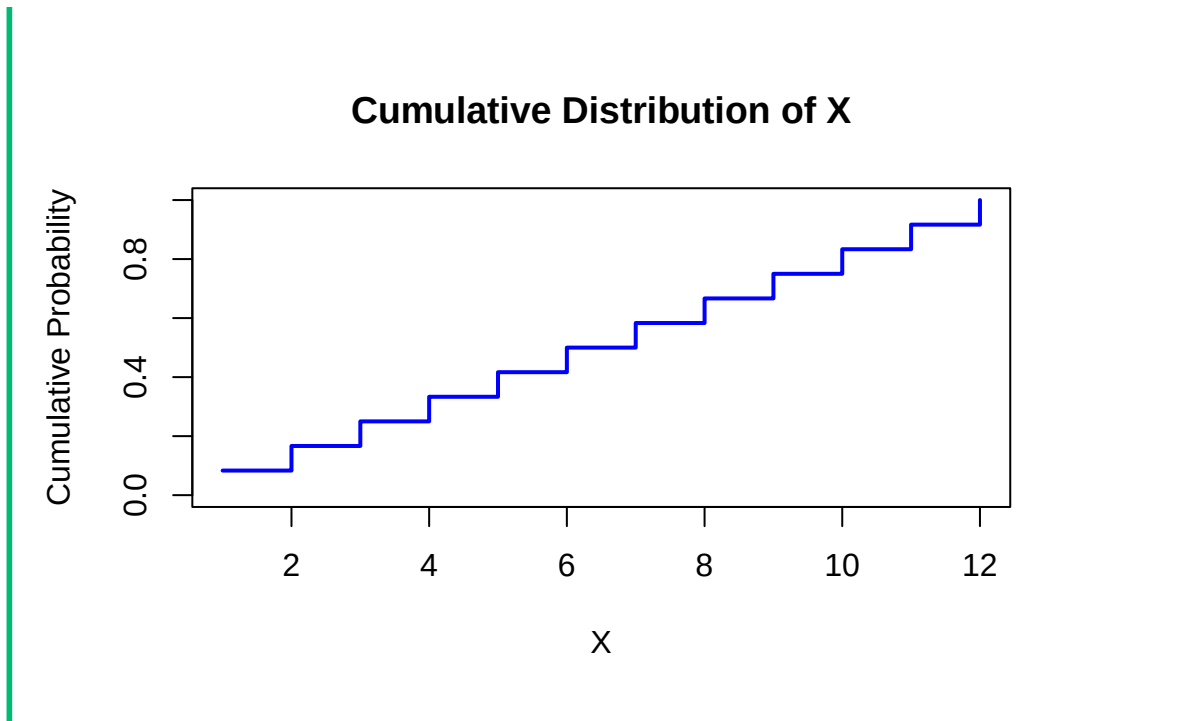
(b) Plot the cumulative probability distribution function $f_X(x)$ in R.

```
# Values of X
x <- 1:12

# Probability of each value
prob <- rep(1/12, 12)

# Calculate cumulative probabilities
cum_prob <- cumsum(prob)

# Plot cumulative distribution
plot(x, cum_prob,
     type = "s",
     main = "Cumulative Distribution of X",
     xlab = "X",
     ylab = "Cumulative Probability",
     col = "blue",
     lwd = 2,
     ylim = c(0, 1))
```



(c) Compute the expected value of X in R.

```
# Values of X
x <- 1:12

# Probability of each value
prob <- rep(1/12, 12)

# Calculate expected value
expected_value <- sum(x * prob)

# Display result
expected_value
```

```
[1] 6.5
```

(d) Compute the variance of X in R.

```
# Values of X
x <- 1:12

# Probability of each value
prob <- rep(1/12, 12)

# Calculate expected value
expected_value <- sum(x * prob)

# Calculate variance
variance <- sum((x - expected_value)^2 * prob)

# Display result
variance

[1] 11.91667
```

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi\sigma}}$$

(a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.)

```
# Define the standard normal density function
f <- function(x) {
  exp(-x^2 / 2) / sqrt(2 * pi)
}

# Test the function at x = 0
f(0)

[1] 0.3989423
```

(b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
# Integrate the standard normal density
result <- integrate(f, lower = -Inf, upper = Inf)$value

# Display the result
result
```

```
[1] 1
```

The integral of the standard normal probability density function from negative infinity to positive infinity is 1.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

This confirms that the total probability under the standard normal distribution is 1.

- (c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
# Calculate the expected value
expected_value <- integrate(
  function(x) x * f(x),
  lower = -Inf,
  upper = Inf
)$value

# Display result
expected_value
```

```
[1] 0
```

The expected value is calculated using:

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

The result from R is 0.

Therefore, the expected value of the standard normal distribution is 0.

- (d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
# Calculate the variance
variance <- integrate(
  function(x) x^2 * f(x),
  lower = -Inf,
  upper = Inf
)$value

# Display the result
variance
```

```
[1] 1
```

The variance is calculated using:

$$\text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

Since the mean is 0, the result from R is:

$$\text{Var}(X) = 1$$

Therefore, the variance of the standard normal distribution is 1.

(e) Compute $\Pr(Z \leq 0.83)$

```
# Calculate P(Z <= 0.83)
probability <- integrate(
  f,
  lower = -Inf,
  upper = 0.83
)$value

# Display result
probability
```

```
[1] 0.7967306
```

The probability is calculated by integrating the standard normal density function from negative infinity to 0.83.

$$P(Z \leq 0.83) = \int_{-\infty}^{0.83} f(z) dz$$

$$P(Z \leq 0.83) \approx 0.7967$$

Therefore, the probability is approximately 79.67%.

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
# Calculate probability between -0.83 and 0.83
probability <- integrate(
  f,
  lower = -0.83,
  upper = 0.83
)$value

# Display result
probability
```

```
[1] 0.5934612
```

The probability is calculated by integrating the standard normal density function between -0.83 and 0.83.

$$P(-0.83 \leq Z \leq 0.83) = \int_{-0.83}^{0.83} f(z) dz$$

$$P(-0.83 \leq Z \leq 0.83) \approx 0.5935$$

Therefore, the probability is approximately 59.35%.

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
# Calculate probability using pnorm()
prob_pnorm <- pnorm(0.83) - pnorm(-0.83)

# Display result
prob_pnorm
```

```
[1] 0.5934612
```

Using the `pnorm()` function:

$$P(-0.83 \leq Z \leq 0.83) = \Phi(0.83) - \Phi(-0.83)$$

$$= 0.7967306 - 0.2032694$$

$$= 0.5934612$$

The result using `pnorm()` is approximately 0.5935. This matches the result from Q6(f), where we used `integrate()`. Therefore, both methods give the same probability.

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \quad 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
# Define the probability density function
f_prison <- function(x) {
  ifelse(x > 0 & x < 3, x^2 / 9, 0)
}

# Test the function at x = 2
f_prison(2)
```

```
[1] 0.4444444
```

The probability density function was defined in R as:

$$f(x) = \frac{x^2}{9}, \quad 0 < x < 3$$

The function returns 0 outside this interval. At $x = 2$, the density is approximately 0.4444.

(b) Integrate $f(x)$ over the interval $[0,3]$.

```
# Integrate the prison sentence density
result <- integrate(
  f_prison,
  lower = 0,
  upper = 3
)$value

# Display the result
result
```

```
[1] 1
```

The probability density function is integrated from 0 to 3.

$$\int_0^3 \frac{x^2}{9} dx = \left[\frac{x^3}{27} \right]_0^3$$
$$= \frac{27}{27} = 1$$

The result is 1, confirming that the total probability over the interval is 1.

(c) Compute the expected prison sentence.

```
# Calculate the expected prison sentence
expected_sentence <- integrate(
  function(x) x * f_prison(x),
  lower = 0,
  upper = 3
)$value

# Display the result
expected_sentence
```

```
[1] 2.25
```

The expected prison sentence is calculated using:

$$E(X) = \int_0^3 x f(x) dx$$
$$E(X) = \int_0^3 \frac{x^3}{9} dx$$

$$E(X) = \left[\frac{x^4}{36} \right]_0^3$$

$$E(X) = \frac{81}{36} = 2.25$$

Therefore, the expected prison sentence is 2.25 years.

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

- (i) Plot the standard normal probability distribution using `curve()`. Specify the range as `[-5,5]` using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".
- (ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.
- (iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.
- (iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.
- (v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

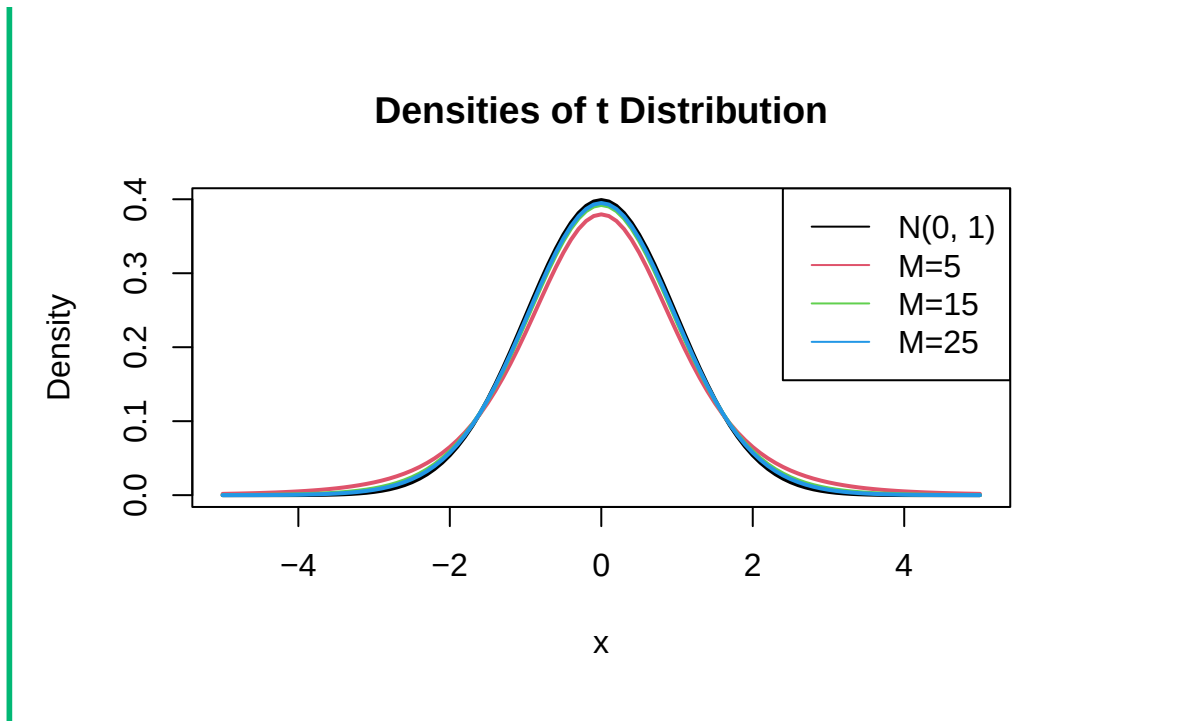
```
# Plot the standard normal distribution
curve(dnorm(x),
      xlim = c(-5, 5),
      ylab = "Density",
      main = "Densities of t Distribution",
      col = 1,
      lwd = 2)

# Student t distribution with 5 degrees of freedom
curve(dt(x, df = 5),
      add = TRUE,
      col = 2,
      lwd = 2)

# Student t distribution with 15 degrees of freedom
curve(dt(x, df = 15),
      add = TRUE,
      col = 3,
      lwd = 2)

# Student t distribution with 25 degrees of freedom
curve(dt(x, df = 25),
      add = TRUE,
      col = 4,
      lwd = 2)

# Add legend
legend("topright",
      c("N(0, 1)", "M=5", "M=15", "M=25"),
      col = 1:4,
      lty = c(1, 1, 1, 1))
```



(b) What do you observe as the degrees of freedom increase?

As the degrees of freedom increase, the Student's t-distribution becomes more similar to the standard normal distribution. The tails become thinner, and the curve becomes slightly taller in the middle. When the degrees of freedom reach 25, the t-distribution is very close to the normal distribution.